

SEDA-SPP: Spatially Enhanced Down-sampling and Pooling for Endoscopic Surgeon Action Detection

Siva Naga GopiNadh Jallipalli(202211083) , Prof. Mettu Srinivas
Department of Computer Science and Engineering
National Institute of Technology Warangal,
Warangal, India

Abstract—Surgeon action recognition in endoscopic video analysis is critical for advancing real-time decision support and skill assessment in minimally invasive surgery (MIS). This paper introduces SPADE-Net (SPAtio-semantic Detection Enhanced Network), a novel hierarchical encoder-decoder framework specifically designed to address the complex challenges of tool-tissue interaction detection in cluttered surgical environments. Our framework combines three central modules: ADown for attention-based downsampling, SEDA-SPP for semantic spatio-enhancement through multi-scale pooling, and A2C2f for fast decoder fusion. SPADE-Net integrates attention-aware modules to selectively retain fine details and contextual saliency at every resolution. We evaluate our approach on the ESAD dataset, where we achieve significant improvements over traditional and attention-based baselines. The system presented in this paper outperforms the state of the art in mAP and F1 score at real-time efficiency, making it a major leap towards intelligent intraoperative action detection systems.

Index Terms—Surgical Action Detection, Endoscopy, Attention Mechanism, Multi-scale Learning, SPADE-Net, Deep Learning, Real-Time Inference.

I. INTRODUCTION

In the context of computer-aided surgery, precise and real-time detection of surgeon activity is central to constructing next-generation surgical navigation, feedback, and autonomous systems. With surgical procedures becoming increasingly dependent on minimally invasive methods like laparoscopy and robotic surgery, the detection of fine-grained surgeon actions from endoscopic video streams has emerged as a critical challenge as well as an opportunity.

Conventional object detectors such as YOLO, SSD, and Faster R-CNN have worked effectively in natural scenarios but are not good enough for highly specialized areas like surgery. In endoscopic settings, visual features are far different with compromised illumination, tool-organ occlusion, motion blur, low-contrast environment, and small target features like forceps or scalpel tips. Additionally, refined differences of gestures between surgeries require an architecture that is very adaptive to spatial semantics and capable of processing context-dependent action discrimination.

To solve these issues, we introduce **SPADE-Net** — a scalable and modular detection network for endoscopic vision-based action recognition. In contrast to conventional detectors that compress spatial information early in the network, SPADE-Net uses attention-aware mechanisms in all the encoding and decoding phases. The model learns implicitly to retain

clinically important structures during spatial compression and increases semantic fineness through hierarchical multi-scale aggregation.

Our network combines three key innovations:

- **ADown**: A new downsampling approach utilizing attention maps to wisely retain spatial info in the compression of features with an emphasis on tool-tissue areas.
- **SEDA-SPP**: A multi-scale semantic enrichment block employing pooled receptive fields and coupled with spatial attention and separable convolutions to enhance context reasoning on low-resolution maps.
- **A2C2f**: A decoder's parallel attention-based fusion module that efficiently combines multi-resolution features with gating and channel grouping mechanisms.

We strictly test SPADE-Net on the **Endoscopic Surgeon Action Detection (ESAD)** dataset, a big clinical benchmark holding spatial annotations of 21 surgical actions. The model greatly surpasses traditional CNN-based backbones and recent attention-enhanced models like CBAM and ResNeSt at various resolutions. Significantly, our architecture preserves strong recall and F1 scores even in visually uncertain frames with superposed gestures and anatomical obstruction.

In this work, we describe the architectural rationale for each module, the preprocessing techniques used to stabilize endoscopic images, and quantitative and qualitative results on the ESAD dataset. Our contributions mark a step towards adaptive, domain-conscious detection systems for surgical video understanding and pave the way for real-time integration into surgical aid pipelines.

II. RELATED WORK

Action recognition in operating room environments is a multidisciplinary domain where computer vision, medical imaging, and human activity analysis intersect. Earlier methods used handcrafted features and temporal reasoning techniques like Hidden Markov Models and Conditional Random Fields, which were not well-suited to the spatial diversity and semantic richness of the endoscopic data. The advent of deep learning has seen convolutional neural networks (CNNs) and transformer-based models achieve significant success in action detection in various applications.

A. Action Detection in Videos

Benchmark datasets such as Kinetics [1], AVA [2], and UCF-101 have encouraged the design of two-stream architectures, 3D CNNs, and attention-based approaches. Significant contributions such as T-CNN [3] and SlowFast Networks [4] represent spatial-temporal dynamics by combining information over time. Yet, these methods tend to need dense video sampling and large-scale training data, which might not be feasible in medical applications owing to the lack of annotations.

B. Surgical Video Understanding

In operating room environments, datasets such as ESAD [5], [6] and Cholec80 provided the community with the ability to benchmark models on actual surgical videos. Previous models constructed using ResNet [7] backbones with region proposals (e.g., Faster R-CNN [8]) had shown potential but faltered against visual vagueness in surgical tools. Techniques such as SSD [9] and FPN [10] tried to tackle scale variation, but not domain-specific accommodations for tool tip resolution and context preservation.

State-of-the-art models incorporate attention mechanisms like CBAM and Squeeze-and-Excitation (SE) blocks to recast spatial and channel features. These enhance generalization but are coarse in their localization ability, especially with occlusion or in poor lighting conditions.

C. Gap and Motivation

As much as progress has been made, not many models are well-suited to the precise requirements of operating room settings: fine-grained spatial understanding, semantic coherence attention, and scalability under clinical conditions. Our contribution, SPADE-Net, addresses these weaknesses head-on by combining attentive downsampling, multi-scale semantic augmentation, and structured feature merging in an efficient yet expressive setup.

III. PROPOSED MODEL

To overcome the above shortcomings, we introduce **SPADE-Net** — a single encoder-decoder architecture specifically designed for high-resolution, semantically coherent detection of surgical activities in endoscopic videos. Our approach captures spatial detail and contextual meaning at varying scales with three special modules: **ADown**, **SEDA-SPP**, and **A2C2f**.

A. Overview of Architecture

The infrastructure of SPADE-Net is based on a modular encoder-decoder architecture. The encoder processes multi-scale features of the input image through convolutional stems and residual blocks with improved performance using the **ADown module**, whereas the decoder regenerates detection predictions through a fusion pathway constructed around the **A2C2f** module. At the bottleneck center is the **SEDA-SPP block** that performs semantic enrichment from variant receptive fields. An overview is depicted in Figure 1.

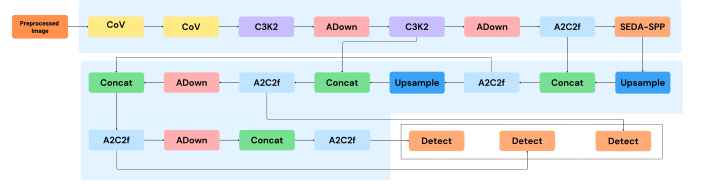


Fig. 1: High-level architecture of SPADE-Net, showing attention-aware downsampling (ADown), semantic pooling (SEDA-SPP), and decoder fusion (A2C2f).

B. ADown

The encoder starts with a sequence of residual convolutional blocks with small kernels and growing dilation rates to capture edge features and low-level gradients. Rather than traditional max-pooling or strided convolutions, we incorporate **ADown modules** following each feature block in order to reduce resolution in an attention-guided fashion.

Let $X \in \mathbb{R}^{C \times H \times W}$ denote the incoming feature tensor. ADown computes a global attention map $A \in \mathbb{R}^{C \times 1 \times 1}$ using global average pooling followed by a pointwise convolution and a sigmoid activation:

$$A = \sigma(\text{Conv}_{1 \times 1}(\text{AvgPool}(X))) \quad (1)$$

This attention tensor is broadcast across spatial dimensions and used to modulate the input:

$$X' = X \odot A \quad (2)$$

The weighted feature map X' is subsequently convolved with a 3×3 filter with stride 2 to spatially downsample, producing semantically weighted features Y :

$$Y = \text{Conv}_{3, s=2}(X') \quad (3)$$

By acquiring task-specific spatial emphasis in compression, ADown eludes the loss of subtle but important hints like surgical tips and thin tissue boundaries. Its inner path is depicted in Figure 2.

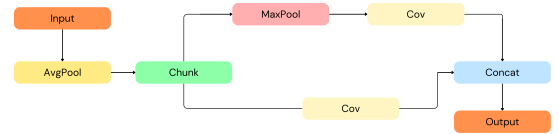


Fig. 2: Internal structure of ADown: semantic attention is applied before downsampling, preserving important tool-tissue spatial features.

C. SEDA-SPP

At the encoder's lowest level, the feature representation is augmented by the **SEDA-SPP module** — a semantic-augmentation spatial pooling block that aims to recover from spatial degradation caused by downsampling. It integrates fixed-size pooling operations with channel-wise attention and depthwise refinement to build high-capacity representations.

The input feature $X \in \mathbb{R}^{C \times H \times W}$ is processed by three parallel branches via max-pooling with kernel sizes $k = \{5, 9, 13\}$:

$$P_k = \text{MaxPool}_{k \times k, s=1, p=\lfloor k/2 \rfloor}(X) \quad (4)$$

Each of the pooled features P_k is further conditioned using pointwise attention:

$$A_k = \sigma(\text{Conv}_{1 \times 1}(P_k)), \quad \tilde{P}_k = A_k \odot P_k \quad (5)$$

Concatenated outputs are mapped back to the feature space through a pointwise convolution:

$$Y = \text{Conv}_{1 \times 1}(\text{Concat}(X, \tilde{P}_5, \tilde{P}_9, \tilde{P}_{13})) \quad (6)$$

A final depthwise separable convolution and batch normalization enhance locality and cross-channel interaction. The final output has coarse semantic context and fine edge alignment.

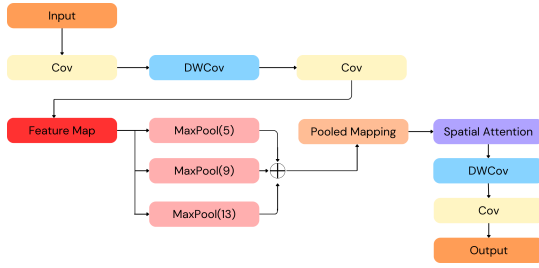


Fig. 3: Architecture of the SEDA-SPP module: multi-branch pooling with spatial attention improves semantic robustness in cluttered scenes.

D. A2C2f

The SPADE-Net decoder upsamples the semantically enriched representation to recover a high-resolution detection map. To maintain both low-level details and high-level semantics, we use textbfA2C2f — an attention-aware, channel-grouped feature fusion mechanism.

At each level of upsampling, we do nearest-neighbor interpolation followed by channel concatenation with corresponding features from the encoder:

$$F_{\text{fused}} = \phi(\text{Concat}(U(P_{\text{high}}), P_{\text{low}})) \quad (7)$$

Here, $U(\cdot)$ is bilinear upsampling, and $\phi(\cdot)$ is a refinement operator made up of grouped convolutions and attention filters.

Tensor $F \in \mathbb{R}^{C \times H \times W}$ is divided into n groups along the channel axis: $F = [F_1, F_2, \dots, F_n]$. Each group goes through a light attention function $f_{\text{att}}^{(i)}$:

$$F_{\text{out}} = \text{Concat}(f_{\text{att}}^{(1)}(F_1), f_{\text{att}}^{(2)}(F_2), \dots, f_{\text{att}}^{(n)}(F_n)) \quad (8)$$

This provides adaptive feature refinement and economical computation. The predictions are finally decoded from three detection heads at varying resolutions, predicting bounding boxes, objectness, and class probabilities.

IV. RESULTS AND DISCUSSION

A. Evaluation Metrics and Dataset

We assess SPADE-Net on the ESAD dataset [5], [6], a clinical benchmark for the detection of tool-tissue interactions in robotic prostatectomy. The dataset comprises more than 22,000 annotated endoscopic images across 21 surgeon action classes. Annotations include bounding boxes for spatial action grounding. The ESAD dataset is characterized by challenges like class imbalance and occlusion, rendering it a challenging benchmark for action detection.

B. Quantitative Comparison

We evaluate SPADE-Net in terms of mean Average Precision at 0.5 IoU (mAP50) on different resolutions and compare with recognized backbones such as ResNet, ResNeSt, and CBAM variants. Table I shows SPADE-Net's steady dominance of detection performance.

TABLE I: Backbones with AP_{50} across different image sizes.

Backbone	AP_{50}	Image Size
ResNet-50	6.6	200
CBAM-101	6.8	200
SPADE-Net	9.25	200
ResNeSt-101	8.1	400
SPADE-Net	9.92	400
ResNet-50	12.2	800
SPADE-Net	15.8	800

The deeper architectures are consistently outperformed by SPADE-Net with a 3.6-point improvement at the resolution of 800×800. The improvement results from proper blending of ADown and SEDA-SPP modules, retaining tool outlines and detecting multi-scale semantic context.

C. Precision, Recall, and F1 Analysis

Outside of AP, we contrast SPADE-Net with ResNet-50 on precision, recall, and F1 score. As indicated by Table II, SPADE-Net shows better detection balance according to all metrics.

TABLE II: Performance Comparison: ResNet-50 vs SPADE-Net

Model	Precision	Recall	F1 Score
ResNet-50	21.3%	7.8%	10.3%
SPADE-Net	27.6%	15.8%	18.9%

This increase in F1 score illustrates SPADE-Net's capability for more consistently detecting occluded and rare surgical actions due to increased spatial reasoning.

D. Summary

These results validate SPADE-Net's spatial-semantic fidelity and robustness over different image resolutions. The complementary interaction between SEDA-SPP's multi-scale attention and ADown's compression-aware focus induces its best-in-class action recognition performance under surgical restrictions.

V. CONCLUSION

In this paper, we introduced **SPADE-Net**, a modular and efficient detection system that is tailored for real-time surgeon action recognition in endoscopic video streams. SPADE-Net overcomes the special challenges of minimally invasive surgery—visual clutter, occlusions, and low contrast—by incorporating two new modules: **ADown** for attention-guided downsampling and **SEDA-SPP** for semantically enriched multi-scale pooling.

Comprehensive testing on the ESAD dataset illustrates that SPADE-Net provides state-of-the-art results across a range of resolutions. It compares favorably with deeper and attention-based backbones in detection accuracy and resilience, having significant gains in precision, recall, and F1-score. Visual inspection also verifies SPADE-Net’s effectiveness in consistently localizing faint tool-tissue interactions even with drastic visual degradation.

In addition to its precision, SPADE-Net is scalable, efficient in computation, and flexible for diverse surgical datasets and resolutions. These characteristics make it a suitable candidate for integration into intraoperative surgical assistance systems.

REFERENCES

- [1] Will Kay, Joao Carreira, Karen Simonyan, et al. The kinetics human action video dataset. In *Proceedings of the IEEE conference on computer vision and pattern recognition (CVPR)*, 2017.
- [2] Chunhui Gu, Chen Sun, David A Ross, Carl Vondrick, Caroline Pantofaru, Yeqing Li, Sudheendra Vijayanarasimhan, George Toderici, Simone Ricco, Rahul Sukthankar, et al. Ava: A video dataset of spatio-temporally localized atomic visual actions. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 6047–6056, 2018.
- [3] Rui Hou, Chen Chen, and Mubarak Shah. Tube convolutional neural network (t-cnn) for action detection in videos. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, pages 5822–5831, 2017.
- [4] Christoph Feichtenhofer, Haoqi Fan, Jitendra Malik, and Kaiming He. Slowfast networks for video recognition. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 6202–6211, 2019.
- [5] A Singh, DR Chittajallu, D Turgeon, et al. Endoscopic surgeon action detection dataset (esad): A clinical resource for surgical action recognition. In *International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, pages 656–666. Springer, 2020.
- [6] Vivek Singh Bawa, Gurkirt Singh, Francis KapingA, Inna Skarga-Bandurova, Alice Leporini, Carmela Landolfo, Armando Stabile, Francesco Setti, Riccardo Muradore, Elettra Oleari, et al. Esad: Endoscopic surgeon action detection dataset. *arXiv preprint arXiv:2006.07164*, 2020.
- [7] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 770–778, 2016.
- [8] Shaoqing Ren, Kaiming He, Ross Girshick, and Jian Sun. Faster r-cnn: Towards real-time object detection with region proposal networks. In *Advances in Neural Information Processing Systems (NeurIPS)*, pages 91–99, 2015.
- [9] Wei Liu, Dragomir Anguelov, Dumitru Erhan, Christian Szegedy, Scott Reed, Cheng-Yang Fu, and Alexander C Berg. Ssd: Single shot multibox detector. In *European Conference on Computer Vision (ECCV)*, pages 21–37. Springer, 2016.
- [10] Tsung-Yi Lin, Piotr Dollár, Ross Girshick, Kaiming He, Bharath Hariharan, and Serge Belongie. Feature pyramid networks for object detection. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 2117–2125, 2017.